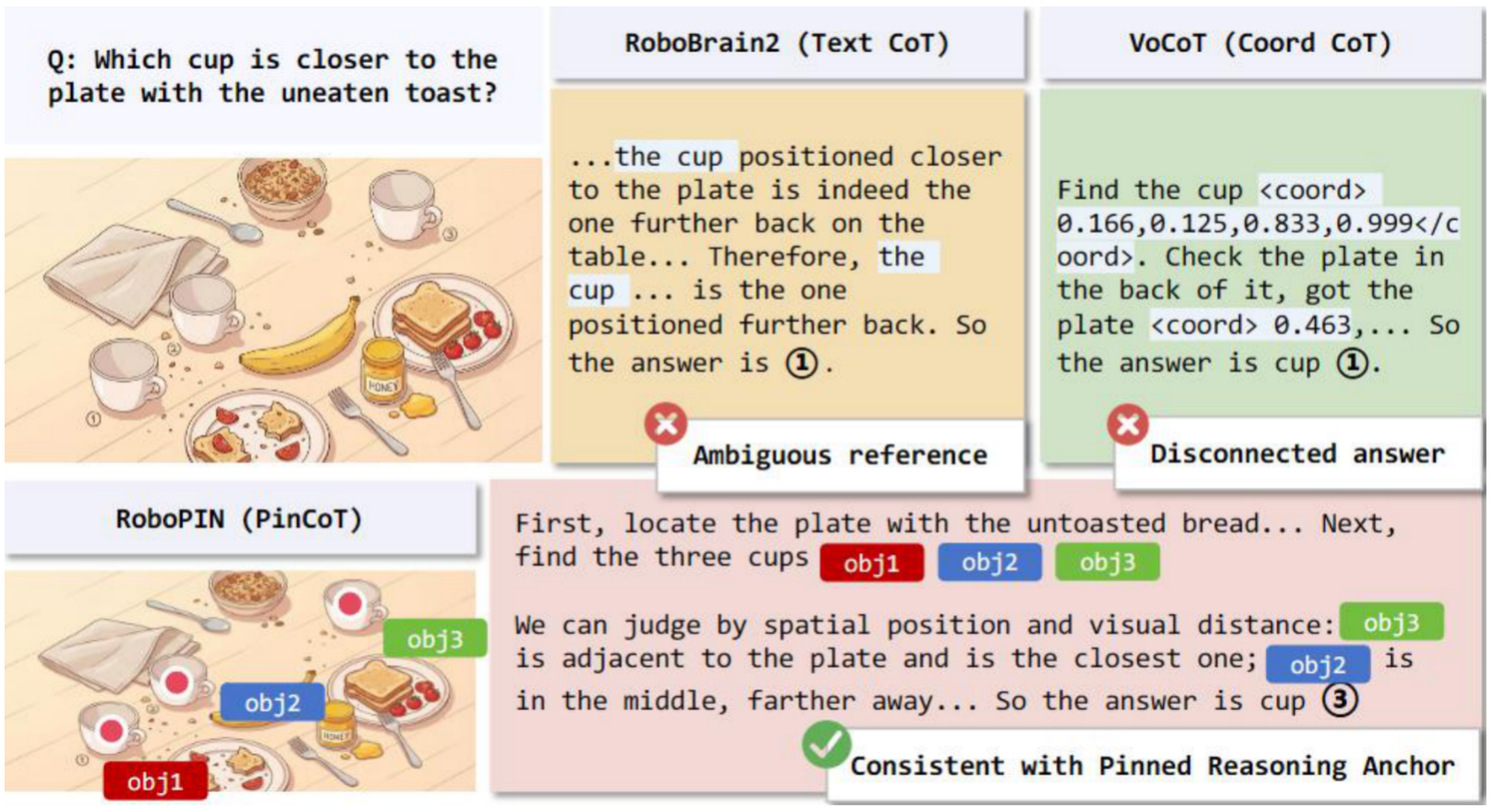
RoboPIN

Grounded Embodied Reasoning via Pinned Chain-of-Thought

Yaoting Huang* · Yifu Yuan*† · Linqi Han · Chengwen Li · Shuoheng Zhang
Xianze Yao · Hongyao Tang · Yan Zheng · Jianye Hao†

Tianjin University · Rio de Janeiro, Brazil · 10–14 November 2026

01 Reason with a persistent visual anchor



THE CORE IDEA

One entity.
One persistent ID.

`a = (name, id, view, point)`

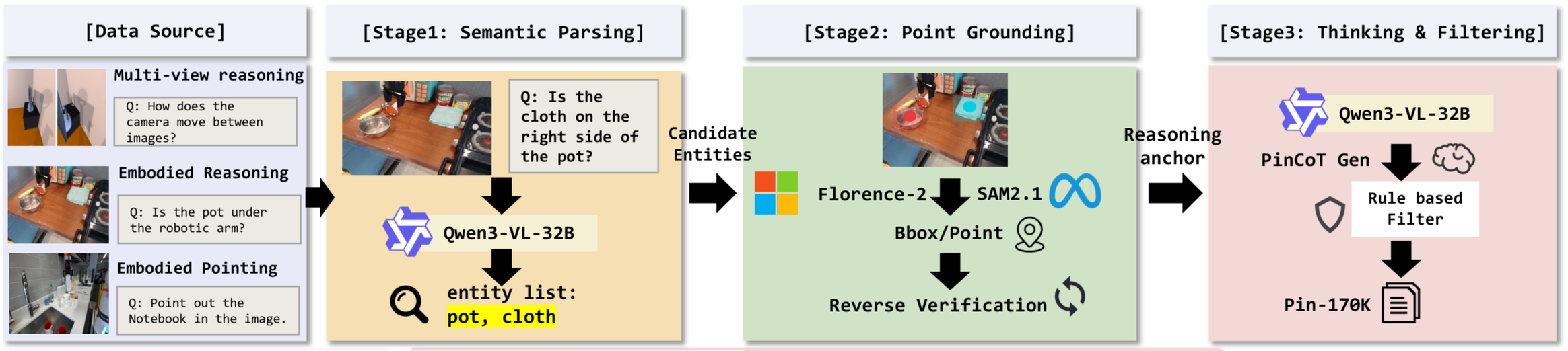
Introduce anchor
name · ID · view · point

↓

Reuse identity
across steps and views

Visual evidence stays traceable.

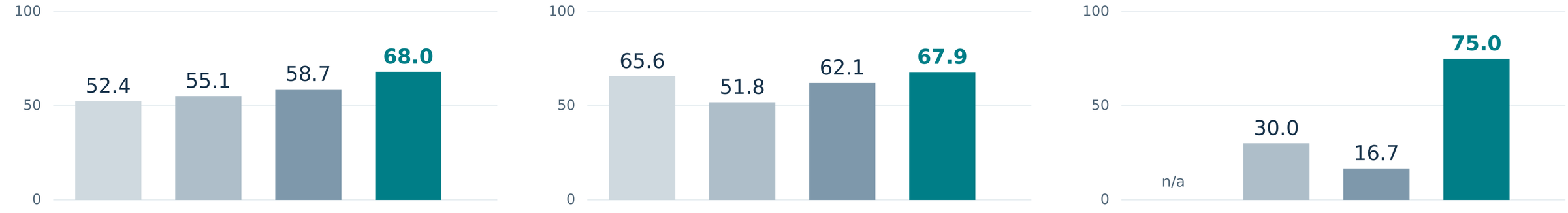
02 How RoboPIN learns to stay grounded



Three-stage post-training

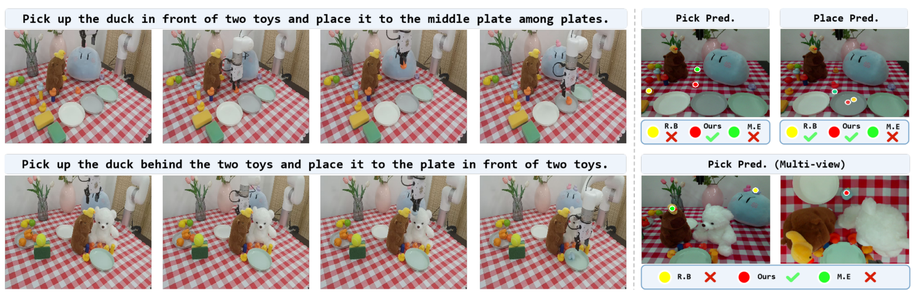
SFT — Embodied knowledge → **CoT-SFT** — Structured reasoning → **RFT / GRPO** — Process alignment

03 Grounding improves embodied reasoning



Scores (%) · Tables 1–3 / Figure 1. Spatial and pointing: arithmetic means. Robot: overall execution success.

04 Reliable execution and consistent entity tracking



Zero-shot xArm execution · Figure 3

Successful executions / 20 per task

20 / 20	14 / 20	11 / 20
Spatial	Multi-view	Long-horizon

18.8% → 99.9%

Cross-step ID coverage · w/o RFT → RoboPIN

PAPER / CODE

QR PLACEHOLDER

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